$$f(t)=\frac{4}{t}-\frac{1}{6t^3}+\frac{8}{t^5}$$

$$\frac{df}{dt}=\frac{df}{dt}4t^{-1}-6t^{-3}+8t^{-5}$$

$$\frac{df}{dt}=-4t^{-2}+18t^{-4}-40t^{-6}$$

$$\frac{df}{dt}=-\frac{4}{t^2}+\frac{18}{t^4}-\frac{40}{t^6}$$